

Table 1: Results of CPLEX and EBP for Small-scale Instances

Instance	CPLEX				GI	EBP							
	P_{lb}	P_{ub}	Gap	$Time(s)$		P_{lb}	P_{ub}	Gap	$label$	$column$	$node$	$avgL$	$Time(s)$
C1_N5_K1	<u>31.58</u>	31.58	0.00%	0.11	31.58	<u>31.58</u>	31.58	0.00%	92	16	1	6	0.06
C2_N5_K1	<u>38.10</u>	38.10	0.00%	0.03	30.12	<u>38.10</u>	38.10	0.00%	159	30	1	5	0.08
C3_N5_K1	<u>30.34</u>	30.34	0.00%	0.03	28.16	<u>30.34</u>	30.34	0.00%	226	21	1	5	0.07
C4_N5_K1	<u>33.62</u>	33.62	0.00%	0.03	25.39	<u>33.62</u>	33.62	0.00%	263	42	1	5	0.06
R1_N5_K1	<u>45.74</u>	45.74	0.00%	0.04	42.36	<u>45.74</u>	45.74	0.00%	356	18	1	7	0.07
R2_N5_K1	<u>35.84</u>	35.84	0.00%	0.03	35.43	<u>35.84</u>	35.84	0.00%	67	22	1	5	0.07
R3_N5_K1	<u>35.04</u>	35.04	0.00%	0.03	30.32	<u>35.04</u>	35.04	0.00%	132	26	1	4	0.07
R4_N5_K1	<u>37.13</u>	37.13	0.00%	0.03	33.50	<u>37.13</u>	37.13	0.00%	108	18	1	4	0.07
RC1_N5_K1	<u>38.06</u>	38.06	0.00%	0.04	37.09	<u>38.06</u>	38.06	0.00%	78	21	1	5	0.07
RC2_N5_K1	<u>17.22</u>	17.22	0.00%	0.09	13.27	<u>17.22</u>	17.22	0.00%	227	33	1	4	0.08
RC3_N5_K1	<u>48.49</u>	48.49	0.00%	0.03	22.70	<u>48.49</u>	48.49	0.00%	103	18	1	5	0.07
RC4_N5_K1	<u>37.48</u>	37.48	0.00%	0.03	37.48	<u>37.48</u>	37.48	0.00%	81	16	1	5	0.08
C1_N5_K2	<u>48.90</u>	48.90	0.00%	0.09	46.08	<u>48.90</u>	48.90	0.00%	143	16	1	4	0.06
C2_N5_K2	<u>57.44</u>	57.44	0.00%	0.06	45.75	<u>57.44</u>	57.44	0.00%	91	28	1	4	0.07
C3_N5_K2	<u>46.39</u>	46.39	0.00%	0.10	37.64	<u>46.39</u>	46.39	0.00%	153	49	1	4	0.07
C4_N5_K2	<u>48.38</u>	48.38	0.00%	0.06	44.34	<u>48.38</u>	48.38	0.00%	245	32	1	4	0.06
R1_N5_K2	<u>59.48</u>	59.48	0.00%	0.13	58.24	<u>59.48</u>	59.48	0.00%	189	29	1	5	0.06
R2_N5_K2	<u>43.98</u>	43.98	0.00%	0.05	41.73	<u>43.98</u>	43.98	0.00%	92	31	1	4	0.09
R3_N5_K2	<u>52.24</u>	52.24	0.00%	0.09	40.68	<u>52.24</u>	52.24	0.00%	209	35	1	4	0.08
R4_N5_K2	<u>53.83</u>	53.83	0.00%	0.09	48.10	<u>53.83</u>	53.83	0.00%	124	25	1	5	0.07
RC1_N5_K2	<u>49.09</u>	49.09	0.00%	0.12	43.41	<u>49.09</u>	49.09	0.00%	125	37	1	4	0.07
RC2_N5_K2	<u>31.49</u>	31.49	0.00%	0.05	26.15	<u>31.49</u>	31.49	0.00%	178	46	1	4	0.06
RC3_N5_K2	<u>77.75</u>	77.75	0.00%	0.10	75.27	<u>77.75</u>	77.75	0.00%	107	21	1	5	0.06
RC4_N5_K2	<u>45.26</u>	45.26	0.00%	0.05	38.93	<u>45.26</u>	45.26	0.00%	111	30	1	4	0.07
C1_N10_K1	<u>90.75</u>	90.75	0.00%	40.08	36.43	<u>90.75</u>	90.75	0.00%	66,218	919	1	6	0.32
C2_N10_K1	<u>89.31</u>	89.31	0.00%	34.86	75.52	<u>89.31</u>	89.31	0.00%	54,891	365	1	8	0.25
C3_N10_K1	<u>113.95</u>	113.95	0.00%	41.57	77.66	<u>113.95</u>	113.95	0.00%	29,924	584	1	8	0.20
C4_N10_K1	<u>79.43</u>	79.43	0.00%	16.81	67.89	<u>79.43</u>	79.43	0.00%	15,110	229	1	8	0.14
R1_N10_K1	<u>67.06</u>	67.06	0.00%	39.86	50.55	<u>67.06</u>	67.06	0.00%	22,231	179	1	8	0.16
R2_N10_K1	<u>62.16</u>	62.16	0.00%	28.49	47.52	<u>62.16</u>	62.16	0.00%	38,218	276	1	7	0.23
R3_N10_K1	<u>76.09</u>	76.09	0.00%	32.85	48.13	<u>76.09</u>	76.09	0.00%	25,231	193	1	8	0.19
R4_N10_K1	<u>47.94</u>	47.94	0.00%	23.56	40.23	<u>47.94</u>	47.94	0.00%	15,781	101	1	7	0.15
RC1_N10_K1	<u>75.94</u>	75.94	0.00%	40.93	62.58	<u>75.94</u>	75.94	0.00%	34,587	108	1	7	0.20

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Table 1 – Continued

Instance	CPLEX				GI		EBP						
	P_{lb}	P_{ub}	Gap	Time(s)	Obj	P_{lb}	P_{ub}	Gap	label	column	node	avgL	Time(s)
RC2_N10_K1	<u>69.54</u>	69.54	0.00%	29.68	69.54	<u>69.54</u>	69.54	0.00%	16,532	105	1	8	0.12
RC3_N10_K1	<u>70.02</u>	70.02	0.00%	33.86	43.57	<u>70.02</u>	70.02	0.00%	41,932	790	1	7	0.28
RC4_N10_K1	<u>63.51</u>	63.51	0.00%	38.98	53.98	<u>63.51</u>	63.51	0.00%	54,165	259	1	7	0.27
C1_N10_K2	<u>121.33</u>	121.33	0.00%	165.78	97.74	<u>121.33</u>	121.33	0.00%	25,771	482	1	5	0.16
C2_N10_K2	<u>118.26</u>	118.26	0.00%	116.70	109.95	<u>118.26</u>	118.26	0.00%	8,737	358	1	7	0.12
C3_N10_K2	<u>140.15</u>	140.15	0.00%	113.29	119.89	<u>140.15</u>	140.15	0.00%	11,671	557	1	6	0.12
C4_N10_K2	<u>114.40</u>	114.40	0.00%	56.56	98.62	<u>114.40</u>	114.40	0.00%	7,663	236	1	6	0.11
R1_N10_K2	<u>100.36</u>	100.36	0.00%	81.43	74.59	<u>100.36</u>	100.36	0.00%	9,983	351	1	6	0.11
R2_N10_K2	<u>95.42</u>	95.42	0.00%	92.83	89.92	<u>95.42</u>	95.42	0.00%	28,836	283	1	7	0.18
R3_N10_K2	<u>115.95</u>	115.95	0.00%	108.37	88.85	<u>115.95</u>	115.95	0.00%	5,187	496	1	5	0.12
R4_N10_K2	<u>68.31</u>	68.31	0.00%	85.10	63.77	<u>68.31</u>	68.31	0.00%	6,101	261	1	6	0.10
RC1_N10_K2	<u>113.51</u>	113.51	0.00%	103.48	83.00	<u>113.51</u>	113.51	0.00%	10,673	483	1	6	0.13
RC2_N10_K2	<u>92.89</u>	92.89	0.00%	110.64	84.07	<u>92.89</u>	92.89	0.00%	10,001	547	1	7	0.16
RC3_N10_K2	<u>99.67</u>	99.67	0.00%	78.08	78.44	<u>99.67</u>	99.67	0.00%	14,491	568	3	6	0.22
RC4_N10_K2	<u>98.28</u>	98.28	0.00%	141.92	79.01	<u>98.28</u>	98.28	0.00%	13,079	451	1	6	0.11
C1_N15_K1	<u>98.13</u>	220.99	125.20%	7200	60.26	<u>98.13</u>	98.13	0.00%	6,629,014	11,285	1	7	22.98
C2_N15_K1	<u>69.47</u>	147.42	112.22%	7200	48.65	<u>69.47</u>	69.47	0.00%	8,665,355	6,204	1	9	16.05
C3_N15_K1	<u>118.19</u>	208.90	76.75%	7200	106.94	<u>118.19</u>	118.19	0.00%	3,447,374	2,955	1	10	7.48
C4_N15_K1	<u>125.87</u>	223.79	77.80%	7200	103.12	<u>126.68</u>	126.68	0.00%	7,383,546	948	1	12	14.69
R1_N15_K1	<u>92.46</u>	162.19	75.43%	7200	71.00	<u>92.46</u>	92.46	0.00%	2,215,205	3,665	1	9	7.07
R2_N15_K1	<u>94.95</u>	179.89	89.46%	7200	72.87	<u>97.18</u>	97.18	0.00%	1,887,114	5,166	1	8	4.53
R3_N15_K1	<u>117.56</u>	225.64	91.94%	7200	94.05	<u>117.56</u>	117.56	0.00%	940,616	10,388	1	8	3.12
R4_N15_K1	<u>113.42</u>	217.42	91.69%	7200	78.31	<u>113.42</u>	113.42	0.00%	3,266,522	38,722	1	9	8.89
RC1_N15_K1	<u>77.16</u>	174.14	125.70%	7200	56.12	<u>77.94</u>	77.94	0.00%	2,410,411	508	1	9	4.74
RC2_N15_K1	<u>116.49</u>	240.03	106.06%	7200	107.15	<u>116.49</u>	116.49	0.00%	2,757,086	1,053	1	9	5.54
RC3_N15_K1	<u>92.25</u>	201.77	118.71%	7200	82.42	<u>95.14</u>	95.14	0.00%	2,260,047	4,712	1	11	4.48
RC4_N15_K1	<u>119.00</u>	251.17	111.07%	7200	107.97	<u>119.00</u>	119.00	0.00%	3,168,829	5,381	1	10	9.29
C1_N15_K2	<u>150.02</u>	233.96	55.95%	5152*	106.55	<u>150.09</u>	150.09	0.00%	482,269	3,416	1	7	1.29
C2_N15_K2	<u>103.64</u>	151.79	46.45%	7200	91.63	<u>103.64</u>	103.64	0.00%	801,977	2,306	1	8	1.56
C3_N15_K2	<u>158.81</u>	219.97	38.51%	3906*	139.88	<u>158.81</u>	158.81	0.00%	279,286	1,315	1	8	0.57
C4_N15_K2	<u>174.54</u>	232.44	33.17%	1963*	133.19	<u>174.82</u>	174.82	0.00%	246,204	1,656	1	9	0.66
R1_N15_K2	<u>122.99</u>	172.65	40.39%	2870*	91.53	<u>124.24</u>	124.24	0.00%	97,448	1,827	1	8	0.38
R2_N15_K2	<u>141.73</u>	187.46	32.26%	7200	104.48	<u>141.73</u>	141.73	0.00%	98,356	1,669	1	7	0.30
R3_N15_K2	<u>160.78</u>	241.03	49.91%	2739*	148.50	<u>162.97</u>	162.97	0.00%	144,365	1,551	1	6	0.50

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Table 1 – Continued

Instance	CPLEX				GI	EBP							
	P_{lb}	P_{ub}	Gap	Time(s)	Obj	P_{lb}	P_{ub}	Gap	label	column	node	avgL	Time(s)
R4_N15_K2	<u>155.13</u>	224.63	44.81%	7200	117.08	<u>155.13</u>	155.13	0.00%	381,096	2,336	1	7	0.97
RC1_N15_K2	<u>124.87</u>	181.97	45.73%	7200	107.30	<u>124.87</u>	124.87	0.00%	292,007	1,341	1	8	0.64
RC2_N15_K2	174.19	256.09	47.02%	4396*	163.22	<u>176.31</u>	176.31	0.00%	291,616	957	1	7	0.59
RC3_N15_K2	<u>134.40</u>	215.62	60.44%	2807*	113.75	<u>134.40</u>	134.40	0.00%	2,222,958	3,823	11	8	3.92
RC4_N15_K2	170.83	266.23	55.84%	3060*	115.25	<u>171.34</u>	171.34	0.00%	251,322	4,348	1	6	0.97
Avg			73.02%	1997				0.00%	710,982	1769	1	7	1.76

Note: The lower bound marked with an underline indicates the optimal solution achieved. The symbol “*” signifies that the solver terminated prematurely due to a memory overflow. “avgL” represents the average route length for each vehicle in the solution.

Table 2: Results of EBP and HBP for Medium-sized Instances

Instance	GI	EBP								HBP					
	Obj	P_{lb}	P_{ub}	Gap	label	column	node	avgL	Time(s)	P_{lb}	ΔP	label	column	Time(s)	ΔT
C1_N20_K2	142.43	<u>189.85</u>	189.85	0.00%	5,408,719	14,201	1	6	15.21	<u>189.85</u>	0.00%	77,470	14,201	1.30	-91.43%
C2_N20_K2	146.58	<u>208.99</u>	208.99	0.00%	8,118,376	20,633	1	5	28.85	207.42	-0.75%	296,940	19,229	1.42	-95.08%
C3_N20_K2	147.21	<u>259.11</u>	259.11	0.00%	2,894,473	18,895	1	7	1448	<u>259.11</u>	0.00%	559,107	18,895	3.29	-99.77%
C4_N20_K2	149.68	<u>208.37</u>	208.37	0.00%	41,577,620	26,990	1	9	597.72	<u>208.37</u>	0.00%	958,904	26,990	4.70	-99.21%
R1_N20_K2	123.19	<u>154.01</u>	154.01	0.00%	2,556,351	14,881	1	8	7.33	<u>154.01</u>	0.00%	808,233	14,881	3.04	-58.50%
R2_N20_K2	144.49	<u>192.96</u>	192.96	0.00%	7,971,982	18,421	1	9	23.44	<u>192.96</u>	0.00%	753,175	18,421	3.05	-86.99%
R3_N20_K2	102.66	<u>161.14</u>	161.14	0.00%	4,456,448	92,968	1	8	14.58	<u>161.14</u>	0.00%	1,528,064	92,968	8.29	-43.18%
R4_N20_K2	137.08	<u>190.13</u>	190.13	0.00%	18,055,150	41,490	1	8	59.93	<u>190.13</u>	0.00%	2,086,905	33,313	8.52	-85.78%
RC1_N20_K2	214.16	<u>269.09</u>	269.09	0.00%	11,276,058	38,211	1	7	36.15	<u>269.09</u>	0.00%	1,139,388	38,211	5.58	-84.57%
RC2_N20_K2	155.15	<u>266.60</u>	266.60	0.00%	9,142,114	33,660	1	7	21.47	<u>266.60</u>	0.00%	1,512,769	32,260	4.82	-77.54%
RC3_N20_K2	186.73	<u>223.90</u>	223.90	0.00%	59,856,553	105,256	1	7	271.78	<u>223.90</u>	0.00%	5,818,533	146,743	23.71	-91.28%
RC4_N20_K2	140.36	<u>170.85</u>	170.85	0.00%	5,151,315	18,873	1	8	13.19	<u>170.85</u>	0.00%	847,374	18,873	4.28	-67.53%
C1_N20_K4	139.27	<u>197.02</u>	197.02	0.00%	17,572,990	8,915	5	5	43.95	<u>197.02</u>	0.00%	28,163	5,696	0.53	-98.79%
C2_N20_K4	160.52	<u>219.70</u>	219.70	0.00%	2,776,287	7,726	1	5	6.81	<u>219.70</u>	0.00%	20,100	4,670	0.35	-94.84%
C3_N20_K4	245.78	<u>286.96</u>	286.96	0.00%	67,929,018	31,658	243	6	114.80	<u>286.67</u>	-0.10%	27,763	6,638	0.45	-99.61%
C4_N20_K4	193.57	<u>233.94</u>	233.94	0.00%	22,275,126	10,120	23	7	54.54	<u>233.94</u>	0.00%	41,855	7,073	0.89	-98.37%
R1_N20_K4	134.77	<u>185.21</u>	185.21	0.00%	382,867	9,656	1	6	1.42	<u>185.21</u>	0.00%	34,577	6,394	0.40	-72.15%
R2_N20_K4	178.25	<u>221.78</u>	221.78	0.00%	6,384,799	8,531	7	7	16.06	<u>221.78</u>	0.00%	30,494	6,338	0.45	-97.21%

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Table 2 – Continued

Instance	GI	EBP								HBP					
	<i>Obj</i>	<i>P_{lb}</i>	<i>P_{ub}</i>	<i>Gap</i>	<i>label</i>	<i>column</i>	<i>node</i>	<i>avgL</i>	<i>Time(s)</i>	<i>P_{lb}</i>	ΔP	<i>label</i>	<i>column</i>	<i>Time(s)</i>	ΔT
R3_N20_K4	127.75	<u>185.13</u>	185.13	0.00%	376,339	9,694	1	6	1.38	<u>185.13</u>	0.00%	25,532	6,079	0.28	-79.42%
R4_N20_K4	172.58	<u>217.84</u>	217.84	0.00%	2,388,238	8,935	3	6	7.89	<u>217.53</u>	-0.14%	33,047	6,049	0.45	-94.24%
RC1_N20_K4	251.15	<u>293.98</u>	293.98	0.00%	38,222,736	164,571	2273	6	68.21	<u>293.98</u>	0.00%	15,655	5,054	0.57	-99.17%
RC2_N20_K4	264.97	<u>298.72</u>	298.72	0.00%	7,020,608	10,088	11	5	12.76	<u>298.51</u>	-0.07%	62,996	5,547	1.05	-91.74%
RC3_N20_K4	226.4	<u>267.35</u>	267.35	0.00%	141,189	6,142	1	5	0.68	<u>267.35</u>	0.00%	25,238	6,142	0.35	-48.37%
RC4_N20_K4	171.73	<u>214.71</u>	214.71	0.00%	728,961	6,624	1	6	1.86	<u>214.71</u>	0.00%	21,980	5,429	0.20	-89.35%
C1_N30_K2	243.88	<u>320.50</u>	320.50	0.00%	150,801,125	2,027,243	1	10	1125	<u>318.36</u>	-0.67%	41,468,278	425,558	276.09	-75.47%
C2_N30_K2	210.22	<u>319.14</u>	319.14	0.00%	60,341,167	439,540	1	8	320.27	<u>319.14</u>	0.00%	41,403,531	439,537	236.16	-26.26%
C3_N30_K2	227.62	<u>347.57</u>	347.57	0.00%	58,746,825	256,929	3	9	250.98	<u>347.55</u>	0.00%	29,161,564	264,236	177.81	-29.15%
C4_N30_K2	249.23	<u>334.09</u>	334.09	0.00%	147,944,714	439,616	1	10	607.69	<u>332.72</u>	-0.41%	47,672,950	339,994	317.69	-47.72%
R1_N30_K2	258.18	<u>296.43</u>	296.43	0.00%	4,878,024	28,675	1	9	15.58	<u>296.43</u>	0.00%	4,275,492	28,675	14.20	-8.85%
R2_N30_K2	221.02	<u>292.37</u>	292.37	0.00%	34,272,936	140,552	1	9	164.84	<u>292.37</u>	0.00%	35,654,143	131,141	142.76	-13.40%
R3_N30_K2	225.36	<u>286.16</u>	286.16	0.00%	8,414,657	65,635	1	8	42.16	<u>286.16</u>	0.00%	7,176,539	65,635	38.33	-9.07%
R4_N30_K2	229.23	<u>294.28</u>	294.28	0.00%	16,282,946	88,430	1	8	84.48	<u>291.38</u>	-0.99%	9,503,896	91,666	63.07	-25.34%
RC1_N30_K2	281.07	<u>364.03</u>	364.03	0.00%	25,763,547	145,337	1	10	108.13	<u>363.09</u>	-0.26%	10,351,028	149,015	59.74	-44.75%
RC2_N30_K2	225.16	<u>302.63</u>	302.63	0.00%	48,360,718	69,434	3	9	146.52	<u>302.00</u>	-0.21%	11,899,750	65,022	47.59	-67.52%
RC3_N30_K2	196.5	<u>264.64</u>	264.64	0.00%	15,951,204	86,131	1	9	65.33	<u>264.64</u>	0.00%	12,010,552	86,131	57.92	-11.34%
RC4_N30_K2	260.25	<u>304.23</u>	304.23	0.00%	43,890,823	144,427	1	9	143.84	<u>301.58</u>	-0.87%	14,774,498	155,379	68.86	-52.13%
C1_N30_K4	318.78	<u>393.00</u>	393.00	0.00%	9,507,122	30,813	1	8	33.71	<u>393.00</u>	0.00%	193,157	18,622	1.67	-95.05%
C2_N30_K4	283.28	<u>399.36</u>	399.36	0.00%	5,708,286	24,141	1	7	17.75	<u>399.36</u>	0.00%	494,437	18,400	3.78	-78.68%
C3_N30_K4	328.14	<u>423.07</u>	423.07	0.00%	13,399,394	29,959	3	8	44.34	<u>423.07</u>	0.00%	267,627	18,205	2.77	-93.75%
C4_N30_K4	298.16	<u>394.26</u>	394.26	0.00%	10,105,745	29,804	1	8	35.82	<u>393.97</u>	-0.08%	158,867	21,369	2.46	-93.12%
R1_N30_K4	282.4	<u>355.65</u>	355.65	0.00%	568,608	20,341	1	8	2.58	<u>355.45</u>	-0.05%	115,282	9,248	1.77	-31.36%
R2_N30_K4	267.31	<u>347.05</u>	347.05	0.00%	4,267,881	28,100	1	7	14.12	<u>347.05</u>	0.00%	332,003	19,473	2.90	-79.43%
R3_N30_K4	266.75	<u>349.07</u>	349.07	0.00%	1,108,465	29,150	1	8	4.68	<u>349.04</u>	-0.01%	216,130	12,691	1.68	-64.10%
R4_N30_K4	295.66	<u>346.36</u>	346.36	0.00%	5,515,374	28,132	1	7	17.58	<u>346.36</u>	0.00%	319,698	20,842	2.93	-83.31%
RC1_N30_K4	349.13	<u>443.21</u>	443.21	0.00%	2,596,441	29,270	1	8	8.80	<u>443.21</u>	0.00%	354,272	18,654	2.35	-73.34%
RC2_N30_K4	272.3	<u>359.56</u>	359.56	0.00%	6,157,985	32,523	1	8	21.43	<u>359.56</u>	0.00%	393,489	21,613	2.66	-87.60%
RC3_N30_K4	250.18	<u>310.94</u>	310.94	0.00%	27,020,634	32,941	7	8	76.51	<u>310.35</u>	-0.19%	374,732	20,978	3.27	-95.72%
RC4_N30_K4	292.56	<u>370.71</u>	370.71	0.00%	4,074,912	35,425	1	7	14.70	<u>370.71</u>	0.00%	194,372	25,840	1.96	-86.65%
C1_N40_K3	420.89	<u>590.64</u>	687.25	16.36%	66,115,718	219,148	1	10	7200	<u>590.64</u>	0.00%	43,527,339	204,896	284.86	-96.04%
C2_N40_K3	376.84	<u>476.38</u>	554.96	16.50%	145,649,686	497,973	1	11	7200	<u>474.27</u>	-0.44%	51,949,751	218,116	436.46	-93.94%
C3_N40_K3	321.88	<u>479.91</u>	569.27	18.62%	71,238,771	148,638	1	9	7200	<u>479.91</u>	0.00%	44,814,161	276,639	334.14	-95.36%
C4_N40_K3	356.82	<u>545.42</u>	643.48	17.98%	41,442,090	138,443	1	10	7200	<u>545.42</u>	0.00%	43,787,302	133,721	266.48	-96.30%

Continued on next page

Table 2 – Continued

Instance	GI	EBP								HBP					
	<i>Obj</i>	<i>P_{lb}</i>	<i>P_{ub}</i>	<i>Gap</i>	<i>label</i>	<i>column</i>	<i>node</i>	<i>avgL</i>	<i>Time(s)</i>	<i>P_{lb}</i>	ΔP	<i>label</i>	<i>column</i>	<i>Time(s)</i>	ΔT
R1_N40_K3	309.96	427.53	494.89	15.76%	24,790,126	95,268	1	10	7200	427.53	0.00%	19,217,944	95,268	121.37	-98.31%
R2_N40_K3	255.83	377.91	451.05	19.35%	40,319,981	112,019	1	10	7200	377.91	0.00%	37,414,000	129,235	263.69	-96.34%
R3_N40_K3	321.38	437.55	437.55	0.00%	42,033,184	113,789	1	10	220.43	437.55	0.00%	12,703,799	113,789	97.43	-55.80%
R4_N40_K3	315.79	425.35	425.35	0.00%	105,966,276	118,626	3	9	600.85	424.43	-0.22%	9,974,568	107,969	78.42	-86.95%
RC1_N40_K3	329.83	441.17	441.17	0.00%	72,424,401	228,050	1	9	687.09	441.17	0.00%	27,763,713	191,138	173.09	-74.81%
RC2_N40_K3	345.29	515.22	515.22	0.00%	29,831,237	186,908	1	8	120.71	515.22	0.00%	14,138,908	186,908	85.44	-29.22%
RC3_N40_K3	335.32	422.83	422.83	0.00%	52,795,069	189,652	1	9	285.95	422.83	0.00%	23,151,466	191,562	183.03	-35.99%
RC4_N40_K3	348.81	446.25	446.25	0.00%	32,225,780	102,642	1	9	132.64	446.25	0.00%	13,902,638	102,642	79.08	-40.38%
C1_N40_K6	536.36	658.94	658.94	0.00%	2,922,865	60,895	3	7	1310	658.94	0.00%	231,121	34,984	4.39	-99.66%
C2_N40_K6	436.43	534.72	534.72	0.00%	960,418	61,326	1	8	351.64	534.72	0.00%	190,059	32,183	3.92	-98.89%
C3_N40_K6	426.73	543.91	543.91	0.00%	1,285,038	52,131	1	7	530.58	543.91	0.00%	229,185	33,360	4.56	-99.14%
C4_N40_K6	484.89	614.05	614.05	0.00%	19,026,313	48,089	1	8	93.72	614.05	0.00%	197,735	30,535	3.56	-96.21%
R1_N40_K6	388.62	471.04	471.04	0.00%	44,775,601	52,979	5	7	176.61	471.04	0.00%	271,397	33,657	4.89	-97.23%
R2_N40_K6	341.58	426.94	426.94	0.00%	43,653,348	50,148	5	7	164.72	426.94	0.00%	182,098	29,752	4.11	-97.50%
R3_N40_K6	413.26	487.53	487.53	0.00%	55,586,925	47,401	19	7	179.84	487.10	-0.09%	255,128	31,511	5.41	-96.99%
R4_N40_K6	395.67	471.86	471.86	0.00%	7,014,162	45,649	1	7	28.53	471.86	0.00%	242,271	29,312	3.36	-88.21%
RC1_N40_K6	395.46	515.88	515.88	0.00%	16,740,072	52,733	1	8	71.45	515.88	0.00%	423,846	40,444	5.06	-92.92%
RC2_N40_K6	474.44	597.51	597.51	0.00%	8,379,001	36,950	1	7	30.16	597.51	0.00%	149,139	25,941	3.02	-90.00%
RC3_N40_K6	396.33	488.19	488.19	0.00%	12,300,034	49,750	1	7	48.93	488.19	0.00%	509,862	28,732	5.17	-89.44%
RC4_N40_K6	406.65	504.39	504.39	0.00%	9,028,436	48,830	1	8	36.79	504.39	0.00%	278,985	32,499	4.39	-88.08%
Avg					27,706,228	107,885	37	8	757.03		-0.33%	8,764,263	74,289	56.44	-83.22%

Table 3: Results of GI and HBP for Large-scale Instances

Instance	GI		HBP			
	<i>Obj</i>	<i>Obj</i>	<i>Improvement</i>	<i>label</i>	<i>column</i>	<i>Time(s)</i>
C1_N50_K6	529.45	649.50	22.67%	864,215	30,802	10.97
C2_N50_K6	561.81	751.85	33.83%	1,450,019	28,839	17.26
C3_N50_K6	484.20	638.14	31.79%	1,577,308	29,404	18.71
C4_N50_K6	519.21	676.20	30.24%	996,224	23,013	11.91
R1_N50_K6	384.10	524.14	36.46%	1,626,366	24,401	15.34
R2_N50_K6	449.01	575.33	28.13%	730,427	23,455	9.04
R3_N50_K6	451.04	561.69	24.53%	1,857,969	25,866	21.66
R4_N50_K6	403.02	518.47	28.65%	3,497,722	30,296	32.72
RC1_N50_K6	468.53	648.99	38.52%	564,242	25,924	8.26
RC2_N50_K6	471.16	628.61	33.42%	2,027,620	33,295	20.03
RC3_N50_K6	447.00	591.15	32.25%	2,294,769	32,149	20.46
RC4_N50_K6	537.93	696.90	29.55%	1,300,456	28,783	13.18
RC1_N80_K8	977.33	1207.78	23.58%	14,234,163	96,208	318.31
RC2_N80_K8	928.81	1126.70	21.31%	17,903,897	99,597	401.19
RC3_N80_K8	879.24	1151.70	30.99%	31,680,959	88,884	758.66
RC4_N80_K8	890.34	1189.66	33.62%	10,862,336	91,561	252.91
RC1_N80_K8	681.83	919.07	34.79%	19,358,637	80,517	431.78
RC2_N80_K8	637.39	866.21	35.90%	46,541,366	81,436	1009
RC3_N80_K8	677.84	881.96	30.11%	38,015,682	79,362	829.48
RC4_N80_K8	661.46	919.80	39.06%	32,696,828	80,844	860.61
RC1_N80_K8	757.29	1033.34	36.45%	11,511,212	95,412	229.35
RC2_N80_K8	751.65	1011.37	34.55%	14,364,479	115,818	275.62
RC3_N80_K8	763.24	1050.01	37.57%	12,058,632	118,150	253.72
RC4_N80_K8	814.58	1051.77	29.12%	15,378,437	98,951	284.38
Avg			31.55%			254.36

Table 4: Results in Different Scenarios

Case 1													
Scenarios	Time slot	without Coordination						with Coordination					
		<i>CO</i>	<i>DO</i>	<i>DR</i>	<i>DT</i>	<i>LD</i>	<i>LT</i>	<i>CO</i>	<i>DO</i>	<i>DR</i>	<i>DT</i>	<i>LD</i>	<i>LT</i>
LLR+ OBP	T1	5	0	20.86	94.05	0	0.00	5	0	21.97	88.15	0	0.00
	T2	6	0	25.38	103.75	1	5.27	6	0	27.19	93.79	0	0.00
	T3	4	0	17.74	81.58	0	0.00	4	0	17.74	81.58	0	0.00
	T4	6	0	21.48	118.84	2	8.47	6	0	22.06	108.39	1	2.66
	T5	6	0	25.96	91.26	0	0.00	6	0	26.52	71.64	0	0.00
	T6	1	0	4.73	23.74	0	0.00	1	0	4.73	23.74	0	0.00
	T7	7	0	31.23	109.94	1	7.51	7	0	32.28	102.79	1	4.61
	T8	2	0	10.62	47.15	0	0.00	2	0	10.62	47.15	0	0.00
	T9	11	0	47.10	157.63	2	8.89	11	0	48.49	149.63	2	8.92
	T10	4	0	19.16	80.86	0	0.00	4	0	19.42	74.15	0	0.00
	T11	3	0	13.99	79.66	0	0.00	3	0	14.43	74.68	0	0.00
	T12	8	0	30.80	126.24	2	20.87	8	0	30.81	125.70	2	20.33
Sum		63	0	269.03	1114.70	8	51.01	63	0	276.27	1041.39	6	36.52

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Table 4 – Continued

Scenarios	Time slot	without Coordination						with Coordination					
		CO	DO	DR	DT	LD	LT	CO	DO	DR	DT	LD	LT
HLR+ OBP	T1	5	0	13.16	79.47	2	39.86	5	0	15.75	72.92	2	26.76
	T2	6	0	13.15	88.61	4	70.21	6	0	16.84	75.55	3	41.35
	T3	4	0	11.29	68.57	2	25.87	4	0	11.88	65.32	2	19.75
	T4	6	0	12.17	101.00	3	67.67	6	0	13.12	90.55	3	58.56
	T5	6	0	21.56	64.64	1	16.97	6	0	22.83	61.09	1	13.42
	T6	1	0	4.73	23.74	0	0.00	1	0	4.73	23.74	0	0.00
	T7	7	0	18.24	85.64	4	71.60	7	0	21.27	80.23	3	55.29
	T8	2	0	9.62	34.95	0	0.00	2	0	9.62	34.95	0	0.00
	T9	11	0	36.78	145.51	4	59.35	11	0	40.17	122.20	4	33.66
	T10	4	0	15.94	62.46	1	10.79	4	0	17.08	51.36	1	2.18
	T11	3	0	9.98	68.94	1	16.94	3	0	10.43	65.50	1	13.50
	T12	8	0	16.47	106.89	5	108.26	8	0	16.59	104.70	5	103.77
	Sum	63	0	183.09	930.42	27	487.52	63	0	200.31	848.11	25	368.24
LLR+ OPP	T1	5	0	20.86	94.05	0	0.00	5	0	21.97	88.15	0	0.00
	T2	5	1	23.58	90.42	0	0.00	6	0	27.19	93.79	0	0.00
	T3	5	0	22.63	82.73	0	0.00	4	0	17.74	81.58	0	0.00
	T4	5	1	19.99	95.62	0	0.00	5	1	20.41	92.84	0	0.00
	T5	7	0	28.04	105.63	0	0.00	7	0	30.30	85.73	0	0.00
	T6	1	0	4.73	23.74	0	0.00	1	0	4.73	23.74	0	0.00
	T7	6	1	29.04	86.24	0	0.00	6	1	29.87	81.74	0	0.00
	T8	3	0	14.88	72.18	0	0.00	3	0	14.88	72.18	0	0.00
	T9	10	1	44.62	139.01	0	0.00	11	0	48.13	148.29	0	0.00
	T10	5	0	21.06	100.01	1	7.75	4	0	19.42	74.15	0	0.00
	T11	3	0	13.99	79.66	0	0.00	3	0	14.43	74.68	0	0.00
	T12	6	2	28.04	87.53	0	0.00	6	2	28.04	87.53	0	0.00
	T13	2	0	9.14	67.60	0	0.00	2	0	9.14	67.60	0	0.00
	Sum	63	6	280.60	1124.42	1	7.75	63	4	286.26	1072.00	0	0.00
HLR+ OPP	T1	3	2	12.61	45.21	0	0.00	3	2	12.89	40.80	0	0.00
	T2	4	4	14.93	66.60	1	16.88	6	2	18.13	68.42	2	36.00
	T3	5	3	19.31	58.09	2	10.65	4	2	18.15	52.00	0	0.00
	T4	6	3	15.76	75.24	1	18.10	5	3	18.83	52.52	0	0.00
	T5	8	1	19.87	96.77	3	82.72	7	2	22.03	58.50	2	13.14
	T6	2	0	8.68	34.82	0	0.00	3	0	13.15	42.65	0	0.00
	T7	4	3	18.01	48.35	0	0.00	4	3	18.85	43.85	0	0.00
	T8	3	2	13.22	61.59	1	7.51	3	2	13.36	58.94	1	4.61
	T9	9	4	40.29	92.44	0	0.00	11	2	47.70	95.99	0	0.00
	T10	7	1	14.27	96.03	4	104.15	5	1	16.92	73.94	2	28.72
	T11	3	1	10.82	54.84	1	3.17	3	1	13.41	47.42	0	0.00
	T12	4	5	16.17	63.52	1	11.52	5	4	17.94	51.15	2	4.41
	T13	5	0	14.06	85.00	2	39.03	4	0	13.37	78.13	2	26.16
	Sum	63	29	218.01	878.50	16	293.73	63	24	244.72	764.31	11	113.04
Case 2													
LLR+ OBP	T1	1	0	5.21	26.39	0	0.00	1	0	5.21	23.08	0	0.00
	T2	2	0	10.60	50.18	0	0.00	2	0	10.60	50.18	0	0.00
	T3	6	0	22.84	119.44	2	16.41	6	0	24.33	110.54	2	10.91
	T4	4	0	17.09	95.04	0	0.00	4	0	17.33	90.67	0	0.00
	T5	1	0	4.45	31.52	0	0.00	1	0	4.45	31.52	0	0.00
	T6	5	0	23.58	88.23	0	0.00	5	0	23.70	75.15	0	0.00
	T7	6	0	27.17	88.67	0	0.00	6	0	27.17	88.67	0	0.00

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Table 4 – Continued

Scenarios	Time slot	without Coordination						with Coordination					
		CO	DO	DR	DT	LD	LT	CO	DO	DR	DT	LD	LT
	T8	8	0	39.68	97.60	0	0.00	8	0	39.85	95.35	0	0.00
	T9	4	0	18.84	89.94	0	0.00	4	0	19.30	81.31	0	0.00
	T10	4	0	17.33	95.58	0	0.00	4	0	18.59	82.75	0	0.00
	T11	5	0	20.20	108.24	1	8.85	5	0	20.40	102.33	1	4.14
	T12	7	0	29.14	93.68	0	0.00	7	0	29.14	93.68	0	0.00
	Sum	53	0	236.14	984.51	3	25.26	53	0	240.08	925.23	3	15.05
HLR+ OBP	T1	1	0	5.21	26.39	0	0.00	1	0	5.21	23.08	0	0.00
	T2	2	0	10.06	32.80	0	0.00	2	0	10.10	31.20	0	0.00
	T3	6	0	7.68	100.09	4	120.70	6	0	12.60	89.84	3	72.50
	T4	4	0	8.26	90.79	2	58.80	4	0	10.04	81.09	2	40.76
	T5	1	0	4.45	31.52	0	0.00	1	0	4.45	31.52	0	0.00
	T6	5	0	19.29	58.93	2	7.26	5	0	19.54	57.06	1	5.26
	T7	6	0	17.06	74.35	3	40.11	6	0	17.06	74.35	3	40.11
	T8	8	0	27.67	72.19	3	64.53	8	0	27.85	87.28	3	60.53
	T9	4	0	12.11	76.60	2	35.08	4	0	13.73	71.73	1	18.50
	T10	4	0	12.72	79.86	2	28.16	4	0	14.03	73.40	1	21.33
	T11	5	0	9.24	96.79	3	73.90	5	0	11.20	89.07	3	55.63
	T12	7	0	18.70	81.24	3	55.67	7	0	20.17	76.05	3	40.10
	Sum	53	0	152.45	821.55	24	484.21	53	0	165.98	785.67	20	354.72
LLR+ OPP	T1	1	0	5.21	26.39	0	0.00	1	0	5.21	23.08	0	0.00
	T2	2	0	10.60	50.18	0	0.00	2	0	10.60	50.18	0	0.00
	T3	4	2	19.44	82.52	0	0.00	5	1	22.27	86.55	0	0.00
	T4	6	0	22.01	115.19	1	17.28	5	0	21.25	89.82	0	0.00
	T5	1	0	4.45	31.52	0	0.00	1	0	4.45	31.52	0	0.00
	T6	5	0	23.58	88.23	0	0.00	5	0	23.70	75.15	0	0.00
	T7	6	0	27.17	88.67	0	0.00	6	0	27.17	88.67	0	0.00
	T8	8	0	39.68	97.60	0	0.00	8	0	39.85	95.35	0	0.00
	T9	4	0	18.84	89.94	0	0.00	4	0	19.30	81.31	0	0.00
	T10	4	0	17.33	95.58	0	0.00	4	0	18.59	82.75	0	0.00
	T11	4	1	18.46	91.86	0	0.00	4	1	18.49	90.66	0	0.00
	T12	8	0	30.89	110.54	1	8.74	8	0	33.34	94.08	0	0.00
	Sum	53	3	237.67	968.22	2	26.02	53	2	244.23	889.12	0	0.00
HLR+ OPP	T1	1	0	5.21	26.39	0	0.00	1	0	5.21	23.08	0	0.00
	T2	2	0	10.06	32.80	0	0.00	2	0	10.10	31.20	0	0.00
	T3	2	4	10.35	41.21	0	0.00	3	3	12.97	44.60	0	0.00
	T4	5	3	15.49	72.14	2	28.62	6	1	20.34	64.23	3	14.70
	T5	3	1	7.92	49.33	1	26.05	2	0	9.88	33.54	0	0.00
	T6	4	2	14.45	74.90	1	23.80	4	1	17.38	49.33	0	0.00
	T7	6	2	21.24	59.95	2	10.76	5	2	19.31	53.68	1	1.88
	T8	7	3	30.32	63.42	2	14.41	7	3	30.32	63.42	2	14.41
	T9	5	2	17.61	72.52	2	19.57	5	2	18.44	67.46	1	10.86
	T10	4	2	17.76	48.39	0	0.00	5	1	19.32	58.96	1	14.24
	T11	4	3	13.75	71.78	1	19.71	3	3	12.84	47.53	0	0.00
	T12	7	3	16.76	92.08	3	72.41	7	3	18.42	85.36	3	55.41
	T13	3	0	12.34	50.17	0	0.00	3	0	12.34	50.17	0	0.00
	Sum	53	25	193.27	755.08	14	215.33	53	19	206.85	672.56	11	111.50
Case 3													
LLR+ OBP	T1	6	0	26.30	90.87	1	0.97	6	0	26.84	86.05	0	0.00
	T2	2	0	11.28	54.48	0	0.00	2	0	11.28	54.48	0	0.00

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Table 4 – Continued

Scenarios	Time slot	without Coordination						with Coordination					
		CO	DO	DR	DT	LD	LT	CO	DO	DR	DT	LD	LT
	T3	6	0	28.34	84.86	0	0.00	6	0	28.36	84.73	0	0.00
	T4	3	0	13.76	70.33	0	0.00	3	0	13.76	70.33	0	0.00
	T5	1	0	5.04	24.55	0	0.00	1	0	5.04	24.55	0	0.00
	T6	3	0	14.31	64.30	0	0.00	3	0	14.31	64.30	0	0.00
	T7	5	0	20.50	100.01	1	4.09	5	0	21.99	87.04	0	0.00
	T8	3	0	14.12	77.50	0	0.00	3	0	14.41	74.00	0	0.00
	T9	6	0	25.31	105.99	0	0.00	6	0	25.52	100.03	0	0.00
	T10	5	0	23.52	79.53	0	0.00	5	0	23.80	75.87	0	0.00
	T11	3	0	15.79	71.63	0	0.00	3	0	16.33	67.59	0	0.00
	T12	3	0	13.58	70.76	0	0.00	3	0	13.66	70.28	0	0.00
	Sum	46	0	211.85	894.81	2	5.06	46	0	215.30	859.25	0	0.00
HLR+ OBP	T1	6	0	16.68	78.41	2	44.07	6	0	17.13	79.88	2	47.01
	T2	2	0	8.41	50.48	1	2.48	2	0	8.83	46.32	0	0.00
	T3	6	0	21.11	72.39	2	24.08	6	0	23.05	65.26	1	9.16
	T4	3	0	10.03	59.83	1	12.05	3	0	10.42	56.52	1	8.52
	T5	1	0	5.04	24.55	0	0.00	1	0	5.04	24.55	0	0.00
	T6	3	0	12.78	42.75	0	0.00	3	0	12.97	40.52	0	0.00
	T7	5	0	11.80	81.84	3	55.98	5	0	14.92	71.68	2	28.79
	T8	3	0	10.52	70.28	1	18.38	3	0	11.15	66.78	1	14.88
	T9	6	0	16.09	92.80	2	55.22	6	0	16.91	84.83	2	45.93
	T10	5	0	17.20	64.58	2	16.26	5	0	18.12	61.77	2	10.64
	T11	3	0	12.90	52.25	1	1.60	3	0	14.09	45.50	0	0.00
	T12	3	0	10.74	60.83	1	8.60	3	0	10.96	59.15	1	6.92
	Sum	46	0	153.31	750.99	16	238.72	46	0	163.59	702.76	12	171.85
LLR+ OPP	T1	6	0	26.29	88.81	0	0.00	6	0	26.84	86.05	0	0.00
	T2	2	0	11.28	54.48	0	0.00	2	0	11.28	54.48	0	0.00
	T3	6	0	28.34	84.86	0	0.00	6	0	28.36	84.73	0	0.00
	T4	3	0	13.76	70.33	0	0.00	3	0	13.76	70.33	0	0.00
	T5	1	0	5.04	24.55	0	0.00	1	0	5.04	24.55	0	0.00
	T6	3	0	14.31	64.30	0	0.00	3	0	14.31	64.30	0	0.00
	T7	4	1	18.16	85.01	0	0.00	5	0	21.99	87.04	0	0.00
	T8	4	0	19.06	78.44	0	0.00	3	0	14.41	74.00	0	0.00
	T9	6	0	25.31	105.99	0	0.00	6	0	25.52	100.03	0	0.00
	T10	5	0	23.52	79.53	0	0.00	5	0	23.80	75.87	0	0.00
	T11	3	0	15.79	71.63	0	0.00	3	0	16.33	67.59	0	0.00
	T12	3	0	13.58	70.76	0	0.00	3	0	13.66	70.28	0	0.00
	Sum	46	1	214.44	878.69	0	0.00	46	0	215.30	859.25	0	0.00
HLR+ OPP	T1	4	2	16.25	50.06	0	0.00	4	2	16.81	47.30	0	0.00
	T2	3	1	15.13	40.53	0	0.00	3	1	15.50	39.10	0	0.00
	T3	5	2	22.32	47.22	0	0.00	6	1	23.85	55.28	1	7.28
	T4	4	1	12.85	77.30	1	24.65	3	1	10.89	47.17	0	0.00
	T5	2	0	8.63	39.81	0	0.00	2	0	8.00	51.79	1	1.17
	T6	3	0	12.78	42.75	0	0.00	3	0	12.97	40.52	0	0.00
	T7	2	3	9.78	45.47	0	0.00	3	2	13.31	48.69	0	0.00
	T8	4	2	15.65	57.75	1	7.30	4	1	14.93	49.28	0	0.00
	T9	5	3	19.32	73.17	1	21.27	5	2	21.72	52.03	0	0.00
	T10	6	2	14.61	96.54	3	73.14	5	2	17.92	66.03	1	18.10
	T11	4	1	11.22	76.33	2	39.76	5	0	15.09	77.21	2	41.52
	T12	3	1	12.24	48.11	0	0.00	2	1	9.03	35.30	0	0.00

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Table 4 – Continued

Scenarios	Time slot	without Coordination						with Coordination					
		<i>CO</i>	<i>DO</i>	<i>DR</i>	<i>DT</i>	<i>LD</i>	<i>LT</i>	<i>CO</i>	<i>DO</i>	<i>DR</i>	<i>DT</i>	<i>LD</i>	<i>LT</i>
	T13	1	0	4.63	34.98	0	0.00	1	0	4.63	34.98	0	0.00
	Sum	46	18	175.43	730.02	8	166.12	46	13	184.63	644.68	5	68.07
Case 4													
LLR+ OBP	T1	4	0	16.85	89.33	0	0.00	4	0	17.16	81.74	0	0.00
	T2	2	0	9.98	70.75	0	0.00	2	0	10.25	70.39	0	0.00
	T3	4	0	16.58	87.87	0	0.00	4	0	17.07	87.33	0	0.00
	T4	7	0	29.73	95.35	0	0.00	7	0	31.07	89.55	0	0.00
	T5	7	0	29.19	125.18	2	11.68	7	0	32.13	99.26	1	2.60
	T6	5	0	20.88	112.78	1	7.15	5	0	21.33	106.23	1	4.60
	T7	8	0	34.61	114.94	2	13.47	8	0	36.24	104.94	1	7.08
	T8	2	0	10.62	56.35	0	0.00	2	0	10.65	50.72	0	0.00
	T9	2	0	11.70	56.41	0	0.00	2	0	11.70	56.41	0	0.00
	T10	4	0	16.99	83.89	0	0.00	4	0	16.99	83.89	0	0.00
	T11	5	0	21.02	99.21	1	2.42	5	0	21.12	93.53	0	0.00
	T12	4	0	17.38	76.86	0	0.00	4	0	18.78	57.95	0	0.00
	Sum	54	0	235.54	1068.92	6	34.72	54	0	244.48	981.94	3	14.28
HLR+ OBP	T1	4	0	12.67	63.27	1	15.55	4	0	13.39	57.20	1	4.82
	T2	2	0	7.76	57.07	1	4.14	2	0	8.72	50.45	0	0.00
	T3	4	0	12.28	74.11	1	19.59	4	0	13.26	68.54	1	14.02
	T4	7	0	24.01	75.74	2	29.44	7	0	26.33	73.48	2	24.92
	T5	7	0	12.13	114.10	4	135.49	7	0	17.13	87.58	4	89.95
	T6	5	0	10.26	97.76	3	64.70	5	0	11.54	88.25	3	58.97
	T7	8	0	22.32	96.84	4	90.74	8	0	23.68	92.63	3	82.56
	T8	2	0	10.06	42.25	0	0.00	2	0	10.50	33.22	0	0.00
	T9	2	0	10.43	40.98	0	0.00	2	0	10.54	39.58	0	0.00
	T10	4	0	12.36	78.59	1	26.17	4	0	12.74	74.37	1	21.95
	T11	5	0	13.47	80.73	2	36.67	5	0	13.90	78.57	2	32.35
	T12	4	0	13.59	61.96	1	11.81	4	0	16.73	48.75	0	0.00
	Sum	54	0	161.34	883.40	20	434.30	54	0	178.46	792.62	17	329.54
LLR+ OPP	T1	4	0	16.85	89.33	0	0.00	4	0	17.16	81.74	0	0.00
	T2	2	0	9.98	70.75	0	0.00	2	0	10.25	70.39	0	0.00
	T3	4	0	16.58	87.87	0	0.00	4	0	17.07	87.33	0	0.00
	T4	7	0	29.73	95.35	0	0.00	7	0	31.07	89.55	0	0.00
	T5	5	2	24.88	89.10	0	0.00	6	1	29.44	80.45	0	0.00
	T6	6	1	23.13	125.50	2	12.00	6	0	22.78	106.04	1	3.46
	T7	8	1	35.64	101.15	0	0.00	7	1	34.04	92.74	0	0.00
	T8	3	0	15.79	72.83	0	0.00	3	0	16.04	65.54	0	0.00
	T9	2	0	11.70	56.41	0	0.00	2	0	11.70	56.41	0	0.00
	T10	4	0	16.99	83.89	0	0.00	4	0	16.99	83.89	0	0.00
	T11	4	1	18.54	76.38	0	0.00	5	0	21.12	93.53	0	0.00
	T12	5	0	19.91	100.15	0	0.00	4	0	18.78	57.95	0	0.00
	Sum	54	5	239.72	1048.71	2	12.00	54	2	246.44	965.56	1	3.46
HLR+ OPP	T1	3	1	11.72	51.78	0	0.00	3	1	11.83	48.19	0	0.00
	T2	2	1	7.41	53.64	1	5.92	3	0	9.34	66.54	1	18.82
	T3	4	1	13.27	62.51	1	9.58	3	1	12.04	45.14	0	0.00
	T4	6	2	23.93	67.06	1	12.54	6	2	25.97	61.26	1	6.74
	T5	5	4	20.60	66.13	1	15.35	5	4	21.53	60.46	1	9.68
	T6	5	4	14.38	93.72	2	49.78	6	3	20.73	70.15	1	17.72
	T7	10	2	41.09	128.06	2	22.38	11	0	44.22	116.49	3	18.19

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Table 4 – Continued

Scenarios	Time slot	without Coordination						with Coordination					
		CO	DO	DR	DT	LD	LT	CO	DO	DR	DT	LD	LT
	T8	4	0	17.05	55.63	1	3.76	2	0	10.50	33.22	0	0.00
	T9	2	0	10.43	40.98	0	0.00	2	0	10.54	39.58	0	0.00
	T10	3	1	12.45	48.59	0	0.00	3	1	12.45	48.59	0	0.00
	T11	4	2	16.62	52.94	0	0.00	4	2	16.62	52.94	0	0.00
	T12	5	1	14.31	75.29	2	30.12	6	0	19.91	71.33	1	22.65
	T13	1	0	4.98	33.92	0	0.00	-	-	-	-	-	-
	Sum	54	19	208.24	830.25	11	149.43	54	14	215.69	713.89	8	93.80
Case 5													
LLR+ OBP	T1	5	0	22.56	97.84	1	0.25	5	0	22.85	90.09	0	0.00
	T2	5	0	25.02	79.07	0	0.00	5	0	25.53	75.24	0	0.00
	T3	6	0	25.04	105.10	1	5.94	6	0	25.92	101.14	1	5.94
	T4	5	0	24.85	78.90	0	0.00	5	0	25.27	76.07	0	0.00
	T5	6	0	25.55	93.51	0	0.00	6	0	26.43	87.25	0	0.00
	T6	2	0	10.73	56.44	0	0.00	2	0	10.73	56.44	0	0.00
	T7	2	0	9.10	66.50	0	0.00	2	0	9.10	66.50	0	0.00
	T8	1	0	5.22	35.78	0	0.00	1	0	5.22	35.78	0	0.00
	T9	4	0	17.63	77.46	0	0.00	4	0	17.67	75.51	0	0.00
	T10	6	0	25.96	104.16	1	2.12	6	0	26.23	104.78	1	3.75
	T11	9	0	30.70	131.47	3	35.71	9	0	32.24	124.30	3	25.18
	T12	4	0	16.19	94.75	0	0.00	4	0	16.91	78.73	0	0.00
	Sum	55	0	238.54	1020.98	6	44.02	55	0	244.09	971.83	5	34.87
HLR+ OBP	T1	5	0	11.94	84.78	3	58.97	5	0	13.86	74.20	3	39.31
	T2	5	0	19.73	63.08	2	12.64	5	0	21.15	56.09	1	2.29
	T3	6	0	13.00	88.27	4	74.68	6	0	14.01	93.08	3	73.68
	T4	5	0	16.84	72.58	2	28.44	5	0	17.86	62.58	2	19.61
	T5	6	0	19.09	71.36	2	25.77	6	0	21.13	64.69	1	12.84
	T6	2	0	9.07	45.25	0	0.00	2	0	9.07	45.25	0	0.00
	T7	2	0	8.48	43.17	0	0.00	2	0	8.48	43.17	0	0.00
	T8	1	0	5.22	35.78	0	0.00	1	0	5.22	35.78	0	0.00
	T9	4	0	11.58	67.53	2	24.30	4	0	12.05	65.58	2	20.40
	T10	6	0	13.88	84.60	3	73.82	6	0	14.12	83.10	3	70.50
	T11	9	0	8.45	121.40	5	204.65	9	0	11.71	111.76	5	165.98
	T12	4	0	13.29	75.45	1	16.23	4	0	14.68	59.61	1	7.41
	Sum	55	0	150.58	853.25	24	519.50	55	0	163.33	794.89	21	412.02
LLR+ OPP	T1	4	1	19.94	84.26	0	0.00	5	0	22.85	90.09	0	0.00
	T2	6	0	27.01	106.82	1	6.17	5	0	25.53	75.24	0	0.00
	T3	5	1	23.16	79.60	0	0.00	5	1	24.04	75.64	0	0.00
	T4	6	0	27.98	84.50	0	0.00	6	0	28.75	79.51	0	0.00
	T5	6	0	25.55	93.51	0	0.00	6	0	26.43	87.25	0	0.00
	T6	2	0	10.73	56.44	0	0.00	2	0	10.73	56.44	0	0.00
	T7	2	0	9.10	66.50	0	0.00	2	0	9.10	66.50	0	0.00
	T8	1	0	5.22	35.78	0	0.00	1	0	5.22	35.78	0	0.00
	T9	4	0	17.63	77.46	0	0.00	4	0	17.67	75.51	0	0.00
	T10	5	1	23.46	92.15	0	0.00	5	1	23.77	84.65	0	0.00
	T11	7	3	29.70	96.04	0	0.00	7	3	29.62	94.78	0	0.00
	T12	7	0	23.31	130.88	2	26.33	7	0	26.91	106.36	1	6.06
	Sum	55	6	242.79	1003.94	3	32.50	55	5	250.62	927.75	1	6.06
HLR+ OPP	T1	2	3	9.99	39.52	0	0.00	3	2	12.69	47.60	0	0.00
	T2	6	2	19.40	92.33	2	47.79	6	1	23.40	63.96	2	12.58

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Table 4 – Continued

Scenarios	Time slot	without Coordination						with Coordination					
		CO	DO	DR	DT	LD	LT	CO	DO	DR	DT	LD	LT
	T3	5	3	18.41	58.57	1	4.77	4	3	16.66	50.80	0	0.00
	T4	6	2	19.03	70.09	2	37.46	6	2	19.89	67.93	2	33.14
	T5	6	2	18.47	72.39	2	35.86	6	2	20.80	64.97	2	21.53
	T6	4	0	12.98	70.61	1	18.76	4	0	13.12	68.34	1	16.49
	T7	2	0	8.48	43.17	0	0.00	2	0	8.48	43.17	0	0.00
	T8	1	0	5.22	35.78	0	0.00	1	0	5.22	35.78	0	0.00
	T9	2	2	8.88	37.80	0	0.00	2	2	8.92	35.85	0	0.00
	T10	5	3	17.79	59.52	2	11.28	5	3	18.49	54.52	1	4.70
	T11	9	3	37.10	107.07	1	5.52	9	3	37.76	101.12	1	2.07
	T12	6	1	11.98	88.94	3	76.04	6	1	15.64	78.78	3	47.59
	T13	1	0	3.77	46.77	0	0.00	1	0	3.87	37.58	0	0.00
	Sum	55	21	191.51	822.56	14	237.48	55	19	204.94	750.40	12	138.10

Note: “CO” and “DO” refer to the numbers of completed orders and delayed dispatch orders in each time slot, respectively.